Required sample size for a mean

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need n to estimate μ within margin e at confidence C.
- Population σ, pilot s, or conservative expected SD is supplied.
- This is planning for precision, not a completed CI.

2 DECISION / METHOD

1. Convert C to z*.
2. Choose σ estimate.
3. Compute $n0=(z^*\sigma/e)^2$.
4. ALWAYS ROUND UP.
5. Apply finite-population adjustment only if separately triggered.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$n_0 = (z^*\sigma/e)^2$

ALWAYS ROUND UP

4 FORMULA SHEET VS WHAT I KNOW

Gives: required-mean sample-size formula.
Still know: how to get z, estimate σ if unknown, and always round up.

5 EXCEL ROUTE

```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*sigma^2)/e^2,0)
```

6 TINY WORKED EXAMPLE

C=.95,σ=10,e=3:
$n0=(1.96*10/3)^2=42.684$, so required n=43.

7 TOP EXAM TRAPS

- Rounding to nearest/down.
- Using full interval width instead of margin e.
- Using t critical in this planning formula.
- Forgetting FPC is a later step.

8 SAY ALOUD

Mean sample size comes from the margin equation and is always rounded up.

9 EXIT CHECK

If the allowed margin halves, by what factor does required n change?

ONE-GLANCE VISUAL ROUTE

